



A WHITE PAPER

Token Spenditure as a Performance Indicator

Why your inference bill is a competitive signal,
not a line item to shrink.

Token spend should be sky high — the cost should stay reasonable

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IFYOUREADONEPAGE,READTHISONE.

The Short Version

Your inference bill is not one number. It is three, and collapsing them into "the AI budget" is why so many organizations throttle the exact signal they should be protecting.

THE THREE NEEDLES

Token Spenditure — the rate at which an organization converts human work into AI work. It measures how intensely you are actually using AI, and how far you are willing to push past old habits to get there.

Token Cost — the size of the resulting bill — a subscription to frontier models, or the capital and operating cost of your own infrastructure.

Token Value — what those tokens produce: decisions made, hours saved, a metric moved. The unit is outcomes per token, never tokens.

The claim. Push Spenditure up, because intensity of use is what agent swarms require. Engineer Cost down through architecture. Measure Value against outcomes. Organizations that feel cost pain and respond by throttling usage are optimizing the wrong variable.

The precedent. Early factories did not ask how to minimize electricity. They asked how much more production it could enable, and optimized efficiency afterwards. Tokens are becoming the same kind of industrial input.

Why it takes a swarm. One chatbot is a demo. The Spenditure that moves a real metric comes from the parts a serial assistant does not have at all — reflection loops, cross-validation, consensus between agents.

The hardware turn. Unified-memory desktop nodes handle local swarms today; the 2028-class machines mark the threshold where frontier-scale reasoning moves entirely on-prem. Marginal cost per query falls toward electricity and amortization.

The risk, stated plainly. Deep cost optimization can entrench the infrastructure owners it was meant to help you compete with. And Value is the easiest of the three numbers to fake — forty billion tokens of agents arguing with each other is a failure state, not a milestone.

What this paper does not solve. Value attribution, the process changes high Spenditure demands, and the path for organizations that cannot build private clusters. Named in the closing section rather than glossed over.

1. Three Terms, Deliberately Kept Apart

Token Spenditure — the rate at which an organization converts human work into AI work. Spenditure is the tell: it measures how intensely you're actually using AI, and how far you're willing to push past old habits and old workflows to get there.

Token Cost — the size of the resulting bill, whether that's a subscription to frontier models or the capital and operating cost of your own infrastructure.

Token Value — what those tokens actually produce: decisions made, hours saved, revenue generated, a metric moved. Note the unit: outcomes per token, not tokens. More tokens does not mean more value — a distinction that matters later in this paper. Jensen Huang has put a number on this at the industry level, describing AI data centers as factories where the input is electrons and the output is tokens, and where a single large facility can turn roughly \$50 billion of build cost into \$300–400 billion of intelligence output a year.

There's a century-old precedent for exactly this shift. Early factories didn't ask how to minimize electricity usage — they asked how much more production electricity could enable, and optimized efficiency afterwards. Nobody ran a successful factory by unplugging machines. Tokens are becoming the same kind of industrial input electricity became: you measure what they enable first, and engineer the unit cost down second. That ordering is the whole argument of this paper.

These are not the same number, and the rest of this paper is about why keeping them separate is the whole point.

2. The Claim

If nobody at your company is generating enough Token Spenditure to notice, you're not innovating — you're consuming, lightly.

But Spenditure and Cost are not the same variable, and most organizations confuse them. Feeling cost pain, they throttle usage: fewer agents, shorter context, tighter limits. Wrong fix — that suppresses Spenditure, which is exactly the signal you want to protect. The right posture is the opposite: push Spenditure up, because intensity of use is what parallel reasoning and agent swarms require, while pushing Cost down through architecture. Local inference, purpose-built small models instead of one expensive generalist, smarter retrieval instead of brute-force context — all of it exists to let Spenditure grow without Cost following it in a straight line.

Get this right and the graph shows rising Spenditure on a flat or falling Cost curve. Get it wrong and you either throttle ambition to protect the bill, or scale ambition and quietly bankrupt the project. A competitor who's already decoupled the two is running

circles you can't see, at a Cost you'd never notice on their books. And here's the uncomfortable part: they aren't winning because their models are smarter. They're winning because they can afford to think a hundred times more.

What does Expenditure look like as an actual number, not a philosophy? A few auditable proxies: total tokens per employee per month; the share of business workflows where at least one agent participates; tokens spent on novel work (analysis, hypothesis testing, design exploration) versus routine work (summarization, boilerplate); and the ratio of agent-to-agent tokens to human-prompted tokens — a rising ratio means real swarm activity, not just chatbot usage. Pick two or three, put them on the same dashboard as Cost, and the framework stops being rhetoric.

3. This Is Already Happening

The intensity shift isn't hypothetical — it's visible wherever agentic work has taken hold. GitHub Copilot alone has generated billions of lines of code across millions of developers. The frontier labs run agents against their own work: Anthropic engineers delegate substantial coding tasks to Claude-based agents, and long-lived autonomous coding sessions are now a product category, not a research demo. Microsoft is rebuilding internal workflows around autonomous agents rather than human-triggered chatbot calls. None of this proves any single organization's Expenditure numbers — but it establishes the trend line: the intensity of AI use is climbing steeply everywhere serious work is being done, and the organizations doing it aren't asking whether to spend tokens. They're asking how to afford spending more of them.

4. The Warning Bell

The economics are shifting, on a timeline shorter than most roadmaps account for. Apple's M7 Ultra, reported via Bloomberg for a 2028 Mac Studio, is being designed to support up to 1.5 TB of unified memory shared between CPU and GPU — workstation and server-hall territory, the kind of machine that sits in an office rack or an on-prem room, not a bag. Apple is reportedly building AI server variants on the same silicon lineage, and notably skipping M6 Pro and M6 Max entirely to fast-track the AI-focused M7 generation. The stepping stones arrive sooner: an M5 Ultra Mac Studio with 768 GB lands in late 2026. The detail matters less than the direction — commodity hardware capable of serious local inference is coming, fast, and from more than one vendor.

That removes the excuse. “We can’t afford to run our own models” stops being true the moment the hardware fits in a server closet instead of a hyperscaler’s data center. The honest caveat: large unified memory makes big models *fit*; sustained swarm throughput, model loading, and energy draw are still real constraints, and “runs

locally” is not yet “runs free.” But marginal cost per query is falling toward electricity and amortization — and you don't have to wait for the 1.5 TB class. Unified-memory desktop nodes in the 128–256 GB range handle local agent swarms today (see Section 8); the 2028-class machines simply mark the threshold where frontier-scale reasoning moves entirely on-prem. Organizations that internalize this now will be running dense swarms locally within a few years, while the ones waiting for the next API price cut are still metering every call.

5. Why It Takes Dozens or Hundreds of Agents, Not One Chatbot

Meaningful gains from agentic AI rarely come from a single well-prompted assistant. The largest gains usually emerge when dozens or hundreds of specialized agents attack a problem from different angles, reconciled by a synthesis layer. **One chatbot is a demo.** Many agents fetching data, testing hypotheses, and cross-checking each other — that's a system that moves a metric. The Spenditure in these architectures comes mostly from the parts a serial chatbot doesn't have at all: reflection loops, cross-validation, and consensus between agents.

Moving a real metric generates real Spenditure, a lot of it. If your Spenditure isn't climbing, your swarm probably isn't big enough to matter. If your Cost climbs at the same rate as Spenditure, your architecture isn't doing its job. A few problems, spread across industries so the pattern reads as general:

- **Heavy transport / fuel efficiency** — shaving a few percent off fleet consumption needs agents continuously testing routing, load, and maintenance timing against live data, not a quarterly script.
- **Pharmaceutical discovery** — parallel agents screening candidates and cross-referencing literature outperform any single research assistant; the search space is too large for serial reasoning.
- **Energy grid balancing** — matching renewable volatility to demand needs constant, distributed forecasting between many local agents, not periodic central calls.
- **Logistics and supply chain routing** — the same swarm logic applied to shipping, containers, and warehouse allocation, where the data never stops moving.
- **Content and recommendation at scale** — genuine personalization needs many lightweight agents reasoning per-user in parallel, not one generic model.

Several of these domains already run continuous algorithmic optimization — route solvers, demand forecasters, recommender models. The distinction matters: those systems optimize within a fixed problem formulation someone wrote down years ago. An agent swarm can reformulate the problem, pull in a data source nobody thought to

connect, and question the assumption underneath the solver. That reformulation work is where the new Expenditure goes, and it's why the token bill in these domains *should* reach the boardroom. If it hasn't, the swarm isn't running yet — you're still optimizing the old model.

6. Proof, Not Theory

The Factory — ten agents, five models, running ideation through delivery on a single workstation-class GPU — was built specifically because cloud-API-scale Cost would have made it uneconomical. The constraint drove the design: each model sized to its task rather than one frontier model for everything; local inference for the high-volume, always-on stages; frontier APIs reserved for the few steps where quality genuinely pays for itself. The pipeline runs continuously, so Expenditure stays high, while Cost sits at hardware amortization plus electricity rather than per-token billing. The system exists in its current form *because of* that pressure — the constraint was the innovation engine, not an obstacle to it.

7. The Risk

Push this framework too hard and it can turn against itself in two ways.

First, decoupling Cost from Expenditure takes real infrastructure — custom silicon, local clusters, proprietary optimization pipelines — and that infrastructure is itself a capital-intensive game. Chase Token Cost down far enough and the advantage concentrates in whoever can afford to build it, tilting the field toward the same handful of infrastructure owners this framework was meant to help you compete with. Optimization, left unchecked, can entrench monopoly rather than break it.

Second, Token Value is the easiest of the three numbers to misrepresent. A model optimized purely to produce more tokens per dollar can look efficient on paper while quietly producing less useful output per token — verbose, redundant, or busywork tokens that inflate Expenditure and depress Cost without moving anything real. “Congratulations, your agents generated forty billion tokens arguing with each other” is a failure state, not a milestone. Measure Value against actual outcomes, not against token count, or the other two numbers will lie for you.

8. Hardware Available Right Now

You don't have to wait for 2028. Unified-memory desktop machines already make local agent swarms economically viable:

- **NVIDIA DGX Spark** — roughly \$3,000–4,700. 128 GB unified CPU+GPU memory, full CUDA stack, runs fully offline; handles 70B models without quantization, and

two units can be paired for 256 GB. Throughput per agent is modest, which sounds disqualifying until you remember the architecture this paper argues for: asynchronous background agents care about task completion over hours, not streaming latency for human eyes. A swarm doesn't need any single agent to be fast — it needs many agents to be always on and nearly free.

- **NVIDIA DGX Station** — the bigger sibling, a proper desktop workstation rather than a hand-sized box. More power, more cost, aimed at teams doing heavier fine-tuning rather than inference alone.
- **A high-end gaming PC (RTX 5090-class)** — more raw GPU throughput per dollar, and doubles as a video/render rig. But no unified memory: capped by VRAM, which rules out the model sizes a unified-memory box handles comfortably.

For organizations serious about Spenditure rather than curious about it, the unified-memory route is the honest bridge. The gaming-PC route makes sense only if the real bottleneck is graphics work rather than agent inference.

9. Two Postures, Side by Side

	Legacy posture (throttled)	Decoupled posture (optimized)
Primary goal	Minimize the inference bill	Maximize AI integration density
Architecture	One frontier model, API calls	Swarms of specialized models + routing, local where it counts
Cost behavior	Scales linearly with ambition	Flat or falling while Spenditure climbs
Failure mode	Low ambition, no transformation	Token inflation — vanity Spenditure (mitigated by outcome tracking)
What the board sees	A cost line to cut	Three needles: Spenditure, Cost, Value

10. The Bottom Line

Three needles: Token Spenditure, Token Cost, Token Value. Rising Spenditure with rising Cost and flat Value means you're spending your way to a result — expensive, and eventually unaffordable. Rising Spenditure with flat or falling Cost, and Value pulling ahead of both, means you've built something that scales its own thinking and pays for itself while doing it.

If you want the whole framework in one line, treat it as a conceptual equation, not literal mathematics:

Organizational Intelligence $\approx$ (Token Spenditure $\times$ Token Value) $\div$ Token Cost

Read Value as outcomes, never as token count — otherwise the equation rewards exactly the busywork it's meant to expose. Push the numerator up. Engineer the denominator down. Everything else in this paper is commentary on those two moves.

Three things this short paper deliberately leaves open, each large enough to deserve its own: how to attribute Token Value to outcomes when effects are lagged and multi-causal; what has to change in workflows, review loops, and trust calibration before high Spenditure is even possible; and how organizations that can't build private clusters achieve the same decoupling through open models, efficient fine-tunes, and shared infrastructure. Naming them is not the same as solving them — but a framework that pretended they were solved would be worth less than one that doesn't.

In ten years, organizations won't be measured by how many employees they have. They'll be measured by how much thinking they can afford. Token Spenditure is simply the first attempt to measure that.

A Note on How This Paper Was Made

This paper was produced by the model it describes: one accountable human Product Owner and an agentic team of several frontier models, handling drafting, critique, and revision across four versions. Independent agent reviews caught the first drafts' weakest claims — an overstated hardware timeline, a metric defined as philosophy rather than as a number — and the accountable human overruled the agents more than once, including on what this paper should leave unsolved.

About This Paper

Jesper Karlsson is a UX and product designer, and founder of Otrobonita AI Labs. He has spent three decades in human-computer interaction, enterprise design systems and software development, most recently building autonomous agentic production systems.

Disagreement is the point. If part of this is wrong — and some of it will be — the author would rather hear it than not.

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